ASHMILA ASHRAF

SOFTWARE ENGINEER

ashmilaashrafofficial@gmail.com | +919074283539 | LinkedIn | Portfolio | Github | LeetCode

PROFESSIONAL SUMMARY

Software Engineer with 2.9 years of experience building secure full-stack web applications for India's 112 Emergency Response System (ERSS) at C-DAC. Sole developer on three production applications - owning the full lifecycle from requirements through multi-state deployment. Strong React (v17-v19) skills, with hands-on backend development experience in Node.js and Spring Boot, and working knowledge of PostgreSQL. Hands-on application security experience: vulnerability assessment, remediation, and security testing aligned with OWASP WSTG standards. Comfortable with Python for problem-solving.

EXPERIENCE

Project Engineer **Jun 2023 - Feb 2026**
Centre for Development of Advanced Computing (CDAC) *Thiruvananthapuram, KL, IN*

- Architected and shipped 3 **full-stack web applications** as the sole developer within India's 112 ERSS project at C-DAC, owning the full lifecycle of each app - requirements, design, implementation, security hardening, and multi-state deployment.
- **Application Security:** Remediated 20+ Critical/High severity findings from internal penetration testing - including auth bypass, IDOR/privilege escalation, stored & reflected XSS, SQL injection, broken session management, missing rate limiting, and CORS misconfiguration - using OWASP WSTG-aligned secure coding patterns; validated each fix via Postman and manual retest.
- Implemented Keycloak-based Single Sign-On with HttpOnly cookies, RSA/AES payload encryption, API-key authentication, role-based access control, and environment-based secrets management.
- Designed and built RESTful APIs for CRUD operations, near-real-time dashboard updates (5-second polling), and frontend integration; integrated internal department APIs for data visualizations with React Leaflet geospatial mapping (OSM tiles) and vehicle route plotting.
- Managed application deployment across Indian state and UT environments; set up and maintained NGINX, PM2, WildFly, and Syslog-ng on RHEL 9.4 servers with consistent configuration across all VMs.
- **VAPT :** Performed security testing on chatbot and web applications - UI input attack testing, API testing via Burp Suite & Postman, source code analysis using Snyk/ Bandit (SAST), and dependency analysis (SCA); authored internal vulnerability assessment reports per OWASP WSTG.
- Designed reusable Syslog-ng configuration templates for Java, Node.js, and Python applications - adopted by multiple development teams across the organization for standardized logging
- Authored comprehensive technical documentation including system architecture diagrams, API specifications, and deployment procedures used for team onboarding and knowledge transfer
- Maintained and supported the ERSS Call Tracing application - bug resolution, issue troubleshooting, and deployments.

KEY PROJECTS (C-DAC)

ERSS Scenario Replay **Jul 2023 - Sept 2024**

- ***Technology:*** *React v18, Node.js, React Chrono, React Leaflet, Material UI, Keycloak, NGINX*
- Developed comprehensive timeline visualization application for end-to-end emergency incident data representation with multimedia integration, nested timelines for multi-vehicle dispatch tracking, geospatial mapping with vehicle route visualization and PDF report generation

ERSS Alarms and Alerts **May 2024 - Oct 2025**

- ***Technology:*** *React v17, Spring Boot 2.6, Java 11, Python 3, PostgreSQL, Syslog-ng, Wildfly 21*
- Transformed partially developed system into production-ready, multi-state real-time log monitoring platform with centralized national dashboard featuring interactive India map.
- Built full-stack solution: React frontend, Spring Boot backend with RESTful APIs, log capturing using Syslog-ng and Python scripts for log processing.
- Centralized Logging System: Syslog-ng server captures logs from diverse technology stack (Java, Node.js, Python applications).
- Implemented comprehensive security hardening post-audit.

ERSS Call Center Performance Monitoring System **Mar 2025 - Dec 2025**

- ***Technology:*** *React v19(Vite), Node.js, Keycloak, Material-UI, Radix UI, Shadcn UI, Lucide react, NGINX*
- Developed a real-time performance and availability monitoring web application for ERSS call taker centers - live agent tracking, threshold-based alerting, real-time status indicators, and analytics dashboards.
- Built comprehensive report pages with multi-criteria filtering and export; integrated REST APIs from internal report team.

SKILLS

- **Programming Languages:** Python, JavaScript
- **Frontend:** React (v17, v18, v19 - Vite), HTML, CSS, API Integration, Material UI, React Bootstrap, React Chrono, React Leaflet, React Redux, Shadcn UI, Radix UI
- **Backend:** Node.js, Spring Boot, RESTful APIs
- **Database:** PostgreSQL, MongoDB, PgAdmin
- **Security:** Burp Suite Pro, Snyk(SAST, SCA), Postman, Bandit, OWASP WSTG
- **DevOps & Infra:** Git, Github, Nginx, WildFly, RHEL, Syslog-ng, AWS, Render, Vercel, Neon
- **Other:** Problem Solving, Self-Directed Learning, Cross-Stack Development

PERSONAL PROJECTS

ShopPOS - Point of Sale Application | Live Demo

- ***Technology:*** *React, Node.js, PostgreSQL, Vercel, Render, Neon, AWS (EC2 , RDS)*
- Full-stack POS system with inventory management, customer management, sales report, cart operations, and billing

Clinic Booking Website | Live Demo

- ***Technology:*** *React, Vercel*
- Responsive clinic appointment booking site with doctor profiles, service listings, and contact form

ACADEMIC PROJECT

SignQuiz **2020**

- ***Technology:*** *Python, Django, TensorFlow, Keras, OpenCV*
- Developed a web-based quiz application for learning Indian Sign Language, integrating webcam-based image capture and a pre-trained CNN model for sign recognition.

EDUCATION

Bachelor of Technology in Computer Science and Engineering **2016-2020**
Malabar Institute of Technology | APJ Abdul Kalam Technological University
CGPA: 8.79

ACHIEVEMENTS

- **Copyright Registration:** ERSS Scenario Replay web application Registered under the Indian Copyright Act, 1957 (Certificate No. SW-2025020784)

LANGUAGES

- English, Malayalam